

Exercises

For the first three exercises let G be a graph with vertices a and b , and let $X \subseteq V(G) \setminus \{a, b\}$ be an a - b separator in G .

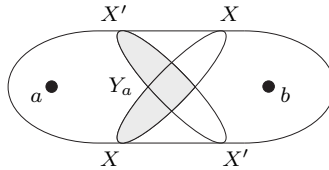
1. ⁻ Show that X is minimal as an a - b separator if and only if every vertex in X has a neighbour in the component C_a of $G - X$ containing a , and another in the component C_b of $G - X$ containing b .
2. ⁻ (continued)
Let $X' \subseteq V(G) \setminus \{a, b\}$ be another a - b separator, and define C'_a and C'_b correspondingly. Show that both

$$Y_a := (X \cap C'_a) \cup (X \cap X') \cup (X' \cap C_a)$$

and

$$Y_b := (X \cap C'_b) \cup (X \cap X') \cup (X' \cap C_b)$$

separate a from b (see figure).



3. (continued)
Are Y_a and Y_b minimal a - b separators if X and X' are? Are $|Y_a|$ and $|Y_b|$ minimum for a - b separators if $|X|$ and $|X'|$ are?
4. Let X and X' be minimal separators in G . Show that if X meets at least two components of $G - X'$, then X' meets all the components of $G - X$ and X meets *all* the components of $G - X'$.
5. ⁻ Deduce the $k = 2$ case of Menger's theorem (3.3.1) from Proposition 3.1.1.
6. ⁻ Prove the elementary properties of blocks mentioned after their formal definition.
7. Show that the block graph of any connected graph is a tree.
8. ⁻ Let G be a k -connected graph, and let xy be an edge of G . Show that G/xy is k -connected if and only if $G - \{x, y\}$ is $(k - 1)$ -connected.
9. (i) Let e be an edge in a 2-connected graph $G \neq K^3$. Show that either $G - e$ or G/e is again 2-connected.
(ii) Does every 2-connected graph $G \neq K^3$ have an edge e such that G/e is still 2-connected?
10. ⁺ Let e be an edge in a 3-connected graph $G \neq K^4$. Show that either $G - e$ or G/e is again 3-connected.

11. Show without using Theorem 3.2.6 that every edge of a 3-connected graph lies on some non-separating induced cycle.
12. Give an inductive proof of Theorem 3.2.6 based on Lemma 3.2.2. You may use the previous exercise.
- 13.⁺ Give an inductive proof of Theorem 3.2.6 based on Lemma 3.2.4.
- 14.⁺ Show that every transitive graph G with $\kappa(G) = 2$ is a cycle.
- 15.⁻ At which point does the first proof of Menger's theorem fail if we assign the contracted vertex v_e to A in G/e only if A in G contains both ends of e , and similarly for B ?
16. When one tries to prove an unknown implication $a \Rightarrow b$, it can be dangerous to attempt to prove $a \Rightarrow c \Rightarrow b$ for some assertion c that clearly implies b : if c is too strong, then $a \Rightarrow c$ may fail even if $a \Rightarrow b$ is true. But the first proof of Menger's theorem appears to be doing just that: it proves the seemingly very strong assertion that, given G, A, B and any edge e of G , we can contract or delete e without decreasing $k(G, A, B)$ – from which the existence of k disjoint A – B paths follows easily by induction. Can one see already at the outset of the proof that this route will in fact not be dangerous?
17. (i) Find the error in the following 'simple proof' of Menger's theorem (3.3.1). Let X be an A – B separator of minimum size. Denote by G_A the subgraph of G induced by X and all the components of $G - X$ that meet A , and define G_B correspondingly. By the minimality of X , there can be no A – X separator in G_A with fewer than $|X|$ vertices, so G_A contains k disjoint A – X paths by induction. Similarly, G_B contains k disjoint X – B paths. Together, all these paths form the desired A – B paths in G .
 (ii) Fill the gap in the proof of (i) by considering, if possible, a vertex or edge outside A and B .
18. Prove Menger's theorem by induction on $\|G\|$, as follows. Given an edge $e = xy$, consider a smallest A – B separator S in $G - e$. Show that the induction hypothesis implies a solution for G unless $S \cup \{x\}$ and $S \cup \{y\}$ are smallest A – B separators in G . Then show that if choosing neither of these separators as X in the previous exercise gives a valid proof, there is only one easy case left to do.
19. Work out the details of the proof of Corollary 3.3.5 (ii).
- 20.⁻ Show that the least number of edges separating two disjoint sets A, B of vertices in a graph G equals the maximum number of edge-disjoint A – B paths in G .
21. Is there a function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that, for every $k \in \mathbb{N}$, every $f(k)$ -edge-connected graph contains between any two vertices k edge-disjoint paths every two of which traverse their common vertices in the same order?

22. Let $k \geq 2$. Show that every k -connected graph of order at least $2k$ contains a cycle of length at least $2k$.
23. Let $k \geq 2$. Show that in a k -connected graph any k vertices lie on a common cycle.
24. Find a subset D of the plane and two infinite subsets $A, B \subseteq D$ such that for every finite set $X \subseteq D$ there is an A – B arc in $D \setminus X$ but D contains no infinite set of disjoint A – B arcs.
- 25.⁺ Prove the following weakening of Gallai's theorem without the help of Mader's theorem. Find a constant c such that, for every k , G and A , either G contains k disjoint A -paths or G has a set of at most ck vertices that meets all its A -paths.
- 26.⁺ Given a collection $\mathcal{S}$ of disjoint vertex sets in a graph $G = (V, E)$, a path in G is an $\mathcal{S}$ -path if it joins distinct sets in $\mathcal{S}$ and has no inner vertices in $S := \bigcup \mathcal{S}$. Show that the following version of Mader's theorem is equivalent to Theorem 3.4.2: The maximum number of disjoint $\mathcal{S}$ -paths in G is always equal to the minimum value of $|V_0| + \sum_{i=1}^r \lfloor \frac{1}{2} |\partial V_i| \rfloor$ taken over all partitions $\{V_0, \dots, V_r\}$ of V such that every $\mathcal{S}$ -path in $G - V_0$ has an edge spanned by some V_i , where ∂V_i is the set of vertices in V_i that lie in S or have a neighbour outside $V_0 \cup V_i$.
27. Derive the edge part of Corollary 3.4.3 from the vertex part.
(Hint. Consider the H -paths in the graph obtained from the disjoint union of H and the line graph $L(G)$ by adding all the edges he such that h is a vertex of H and $e \in E(G) \setminus E(H)$ is an edge at h .)
- 28.[−] To the disjoint union of the graph $H = \overline{K^{2m+1}}$ with k copies of K^{2m+1} add edges joining H bijectively to each of the K^{2m+1} . Show that the resulting graph G contains at most $km = \frac{1}{2} \kappa_H(G)$ independent H -paths.
29. Find a bipartite graph G , with partition classes A and B say, such that for $H := G[A]$ there are at most $\frac{1}{2} \lambda_H(G)$ edge-disjoint H -paths in G .
- 30.[−] Deduce Gallai's theorem, as well as an edge analogue with the same bound of $2k - 2$, from Mader's theorem via Corollary 3.4.3.
31. Derive a suitable version of Menger's theorem from Mader's theorem.
- 32.⁺ Derive Tutte's 1-factor theorem (2.2.1) from Mader's theorem.
(Hint. Extend the given graph G to a graph G' by adding, for each vertex $v \in G$, a new vertex v' and joining v' to v . Choose $H \subseteq G'$ so that the 1-factors in G' correspond to the large enough sets of independent H -paths in G' .)
- 33.[−] Show that $2k$ -edge-connected graphs are k -edge-linked in the sense that for all distinct vertices $s_1, \dots, s_k, t_1, \dots, t_k$ there are edge-disjoint paths $P_i = s_i \dots t_i$ for $i = 1, \dots, k$.
- 34.[−] Show that k -linked graphs are $(2k - 1)$ -connected. Are they even $2k$ -connected?

35. For every $k \in \mathbb{N}$ find an $\ell = \ell(k)$, as large as possible, such that not every ℓ -connected graph is k -linked.
36. Show that if G is k -linked and $s_1, \dots, s_k, t_1, \dots, t_k$ are not necessarily distinct vertices such that $s_i \neq t_i$ for all i , then G contains independent paths $P_i = s_i \dots t_i$ for $i = 1, \dots, k$.
37. Go through the proof of Theorem 3.5.3 monitoring the use of $\|V \setminus X\|^+$. How would the proof fail if $\|G[V \setminus X]\|$ was used instead? Which arguments would become simpler?
38. In the informal discussion preceding the proof of Theorem 3.5.3 we noted that, by Corollary 11.2.3, we cannot expect to find in G a subgraph H that satisfies the premise of Lemma 3.5.4. But then the proof of (*) does find such a subgraph H . Can you explain this?
39. Use Theorem 3.5.3 to show that the function h in Lemma 3.5.1 can be chosen as $h(r) = cr^2$, for some $c \in \mathbb{N}$.

Notes

Although connectivity theorems are doubtless among the most natural, and also the most applicable, results in graph theory, there is still no monograph on this subject. The most comprehensive sources to date are A. Schrijver, *Combinatorial optimization*, Springer 2003, and A. Frank, *Connections in combinatorial optimization*, Oxford University Press 2011. Some areas are covered in B. Bollobás, *Extremal Graph Theory*, Academic Press 1978, in R. Halin, *Graphentheorie*, Wissenschaftliche Buchgesellschaft 1980, and in A. Frank's chapter of the *Handbook of Combinatorics* (R.L. Graham, M. Grötschel & L. Lovász, eds.), North-Holland 1995. A survey specifically of techniques and results on minimally k -connected graphs (see below) is given by W. Mader, On vertices of degree n in minimally n -connected graphs and digraphs, in (D. Miklós, V.T. Sós & T. Szőnyi, eds.) *Paul Erdős is 80*, Vol. 2, Proc. Colloq. Math. Soc. János Bolyai, Budapest 1996.

Theorem 3.2.3 is often attributed to Barnette and Grünbaum (1969). It can also be extracted from W.T. Tutte, *Connectivity in graphs*, Oxford University Press 1966. Tutte's *wheel theorem*, proved in W.T. Tutte, A theory of 3-connected graphs, *Nederl. Akad. Wet. Proc. Ser. A* **64** (1961), 441–455, differs from our Theorem 3.2.5 as follows. As an alternative to the contraction of an edge in the reduction step, the wheel theorem also allows its deletion. Of edges to be contracted, however, it requires that they do not lie in any triangle. The starting set for the construction of all 3-connected graphs therefore consists of all wheels, not only K^4 .

Tutte's wheel theorem has been extended to 3-connected graphs H other than K^4 : from any 3-connected graph $G \succcurlyeq H$ that is not a wheel we can obtain H by contracting or deleting edges step by step, remaining 3-connected at every step. (As in Tutte's theorem, one is not allowed to contract edges that lie in a triangle.) This was proved by S. Negami, A characterization of 3-connected graphs containing a given graph, *J. Comb. Theory, Ser. B* **32** (1982),